

Conditional Statements

- if, else
- else if
- ternary operator
- switch



if-else

condition

```
if(age >= 18) {  
    cout << "can vote";  
} else {  
    cout << "can't vote";  
}
```



Practice Qs

Print the largest of 2 numbers.

10

$\frac{a}{\downarrow}$
5

$\frac{b}{\downarrow}$
10

```
if (a > b) {  
    cout << a  
}
```

Print if a number is Odd or Even.



code.cpp

code.cpp > main()

```

3
4  int main() {
5      int a, b;
6      cout << "enter a : ";
7      cin >> a;
8      cout << "enter b : ";
9      cin >> b;
10
11     if(a >= b) {
12         cout << "a is larger : " << a << endl;
13     } else {
14         cout << "b is larger : " << b << endl;
15     }
16
17     return 0;
18 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
enter a : █



Practice Qs

Print the largest of 2 numbers.

Print if a number is Odd or Even.

num = 5, 10, 3, 18
 ↓ ↓ ↓ ↓
 odd even odd even

even numbers → 2x

2, 4, 6, 8, 10, ...



else if Statement

```
if(marks >= 90) {  
    cout << "A";  
} else if(marks >= 80) {  
    cout << "B";  
} else {  
    cout << "C";  
}
```

condition 2



code.cpp

code.cpp > main()

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int marks = 85;
6
7      if(marks >= 90) {
8          cout << "A" << endl;
9      } else if(marks >= 80) {
10         cout << "B" << endl;
11     } else {
12         cout << "C" << endl;
13     }
14
15     return 0;
16 }

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
B
● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
A
○ apnacollege@Amans-MacBook-Pro Cpp Codes %

```



code.cpp

code.cpp > main()

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int marks = 75;
6
7      if(marks >= 90) {
8          cout << "A" << endl;
9      } else if(marks >= 80) {
10         cout << "B" << endl;
11     } else {
12         cout << "C" << endl;
13     }
14
15     return 0;
16 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
• apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
B
• apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
A
• apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
C
• apnacollege@Amans-MacBook-Pro Cpp Codes %
```



code.cpp x

code.cpp > main()

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int marks = 95;
6
7      if(marks >= 90) {
8          cout << "A" << endl;
9      } else if(marks >= 80) {
10         cout << "B" << endl;
11     } else {
12         cout << "C" << endl;
13     }
14
15     return 0;
16 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
A
B
● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
A
○ apnacollege@Amans-MacBook-Pro Cpp Codes %
```



Practice Qs

income ↓

Create an Income Tax Calculator.

income < 5L

0% tax

income between 5-10L

20% tax

income > 10L

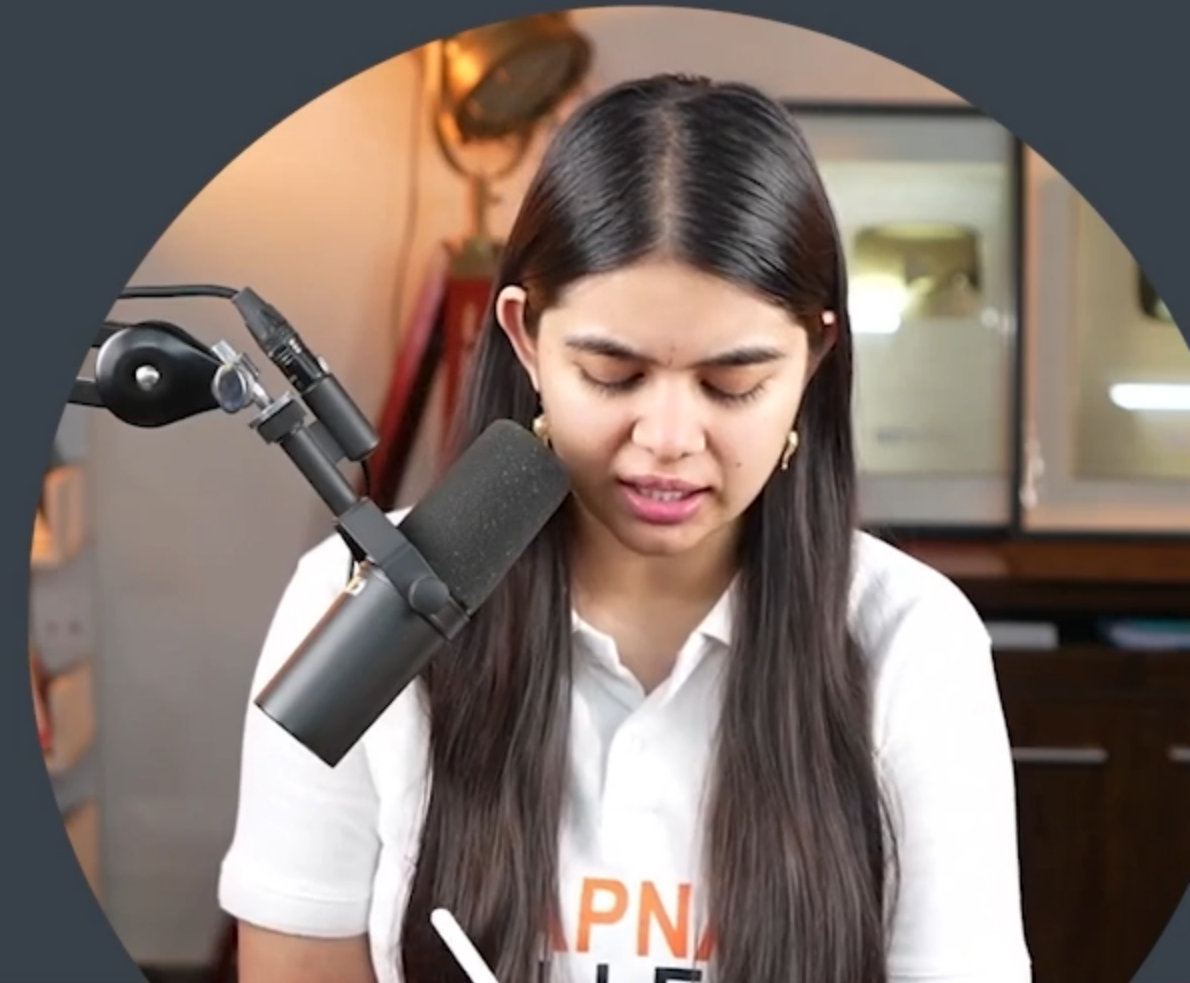
30% tax



Practice Qs

Print the largest of 3 numbers.

↓ ↓ ↓
a b c

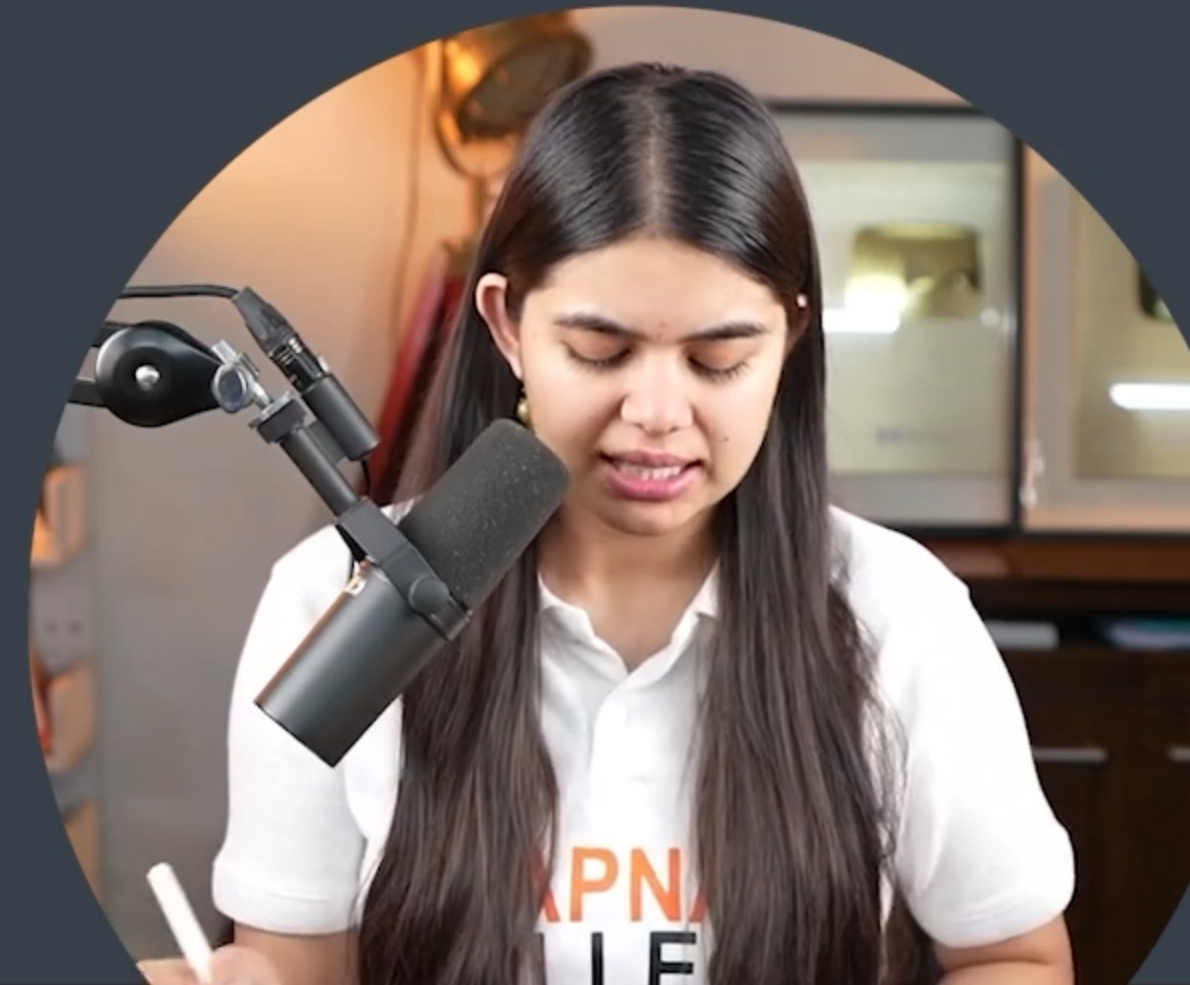


Ternary Operator



variable = condition? statement1 : statement2;

```
bool isAdult = (age >= 18) ? true : false;
```



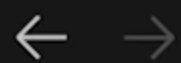
Ternary Operator

variable = condition? statement1 : statement2;



```
bool isAdult = (age >= 18) ? true : false;
```





Cpp Codes



code.cpp

code.cpp > main()

```
3
4  int main() {
5      bool isAdult; //1, 0
6      int age;
7      cout << "enter age : ";
8      cin >> age;
9
10     if(age >= 18) {
11         isAdult = true;
12     } else {
13         isAdult = false;
14     }
15
16     isAdult = age >= 18 ? true : false;
17     return 0;
18 }
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

apnacollege@Amans-MacBook-Pro Cpp Codes %



Switch Statement

```
switch(day) {  
    case 1: cout<<"Monday";  
            break;  
    case 2: cout<<"Tuesday";  
            break;  
    //more cases for 3-7  
    default: cout<<"Invalid";  
}
```



Practice Qs

Build a calculator using Switch for the 4 basic arithmetic operations (+, -, *, /)

input: a, b
character → symbol (+, -, *, /)



code.cpp x

code.cpp > main()

```

27
28 //Switch
29 switch(op) {
30     case '+' : cout << "a + b = " << (a+b) << endl;
31                break;
32     case '-' : cout << "a - b = " << (a-b) << endl;
33                break;
34     case '*' : cout << "a * b = " << (a*b) << endl;
35                break;
36     case '/' : cout << "a / b = " << (a/b) << endl;

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
enter a : 6
enter b : 3
enter operator : +
a + b = 9
● apnacollege@Amans-MacBook-Pro Cpp Codes % g++ code.cpp && ./a.out
enter a : 6
enter b : 3
enter operator : -
a - b = 3
○ apnacollege@Amans-MacBook-Pro Cpp Codes %

```



Question 2 :

```
int main() {  
    int year;  
  
    cin >> year;  
  
    if (year % 400 == 0) {  
        cout << year << " is a leap year.\n";  
    } else if (year % 100 == 0) {  
        cout << year << " is NOT a leap year.\n";  
    } else if (year % 4 == 0) {  
        cout << year << " is a leap year.\n";  
    } else {  
        cout << year << " is NOT a leap year.\n";  
    }  
  
    return 0;  
}
```

Question 5 :

```
int main() {  
    int n;  
    cout << "enter a 3 digit number : ";  
    cin >> n;  
  
    int num = n; //creating a copy  
  
    int dig1 = num % 10;  
    num /= 10;  
    int dig2 = num % 10;  
    num /= 10;  
    int dig3 = num;  
  
    int cubeSum = dig1*dig1*dig1 + dig2*dig2*dig2 + dig3*dig3*dig3;  
  
    if (cubeSum == n) {  
        cout << n << " is an Armstrong Number\n";  
    } else {  
        cout << n << " is NOT an Armstrong Number\n";  
    }  
  
    return 0;  
}
```


Question 6 :

typedef keyword in C++ is used for aliasing existing data types, user-defined data types, and pointers to a more meaningful name.

Eg : *typedef int myInt*

This allows us to declare new integers using myInt, instead of int.

Macros can be of any type. Macros can even be any code block containing statements, loops, function calls etc. They are expanded by the preprocessor before compilation takes place.

const keyword is used to define the constant value that cannot change during program execution.